Comparing High School Accessibility in California and New York

A GIS-Based Exploratory Assessment

CIES 2026 | Jialing Wu (The Ohio State University) | Jialin Yan (University of Rochester)





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Theoretical Lens: Spatial Justice

Spatial justice theory (Soja, 2010) argues that inequity is not just about how many resources exist, but where they are located. The geography of services directly shapes who can access them — and who cannot.

What Prior Research Found

Brazil (2024): School closures reshape the spatial ecology of education access across U.S. cities.

Owens & Candipan (2019): Schools in high- vs. low-income neighborhoods show stark spatial inequality in opportunity.

Planey et al. (2023): Structural factors create racial/ethnic inequities in travel times to acute care hospitals in the rural South.

How GIS Complements The Work

1. Prior studies document inequality through statistics. GIS adds a spatial layer — showing not just that disparities exist, but exactly where they concentrate.

2. By overlaying population data (age, race) onto school locations, it connects supply with demand at a local level.

3. It also reveals whether underserved areas are scattered or clustered — a distinction that matters for policy but sometimes invisible in summary statistics.

This pilot study asks:

How do high schools in California State and New York State differ in their spatial distribution and accessibility?

Research Pipeline

DATA COLLECTION

NCES School Files
2021–22 public & private school universe

Census Boundaries
County & Block Group (BG) shapefiles

ACS 5-Year Data
Population aged 15–18 by gender & race

DATA PROCESSING

Filter to CA & NY → Identify high schools (name contains “High”) → Geocode with Lat/Lon → Reproject all layers

ANALYSIS

County-Level Analysis

1. Spatial join schools to county boundaries
2. Count high schools per county
3. Classify tiers: 0–10 / 11–50 / >50
4. Join ACS youth population → compute ratio
5. Flag underserved: >2,000 youth per school

Block Group Accessibility (20km)

1. Compute BG centroids as demand points
2. Build spatial index on school locations
3. Count schools within 20km of each BG
4. Classify: 0 schools vs. ≥20 schools

County-Level School Distribution

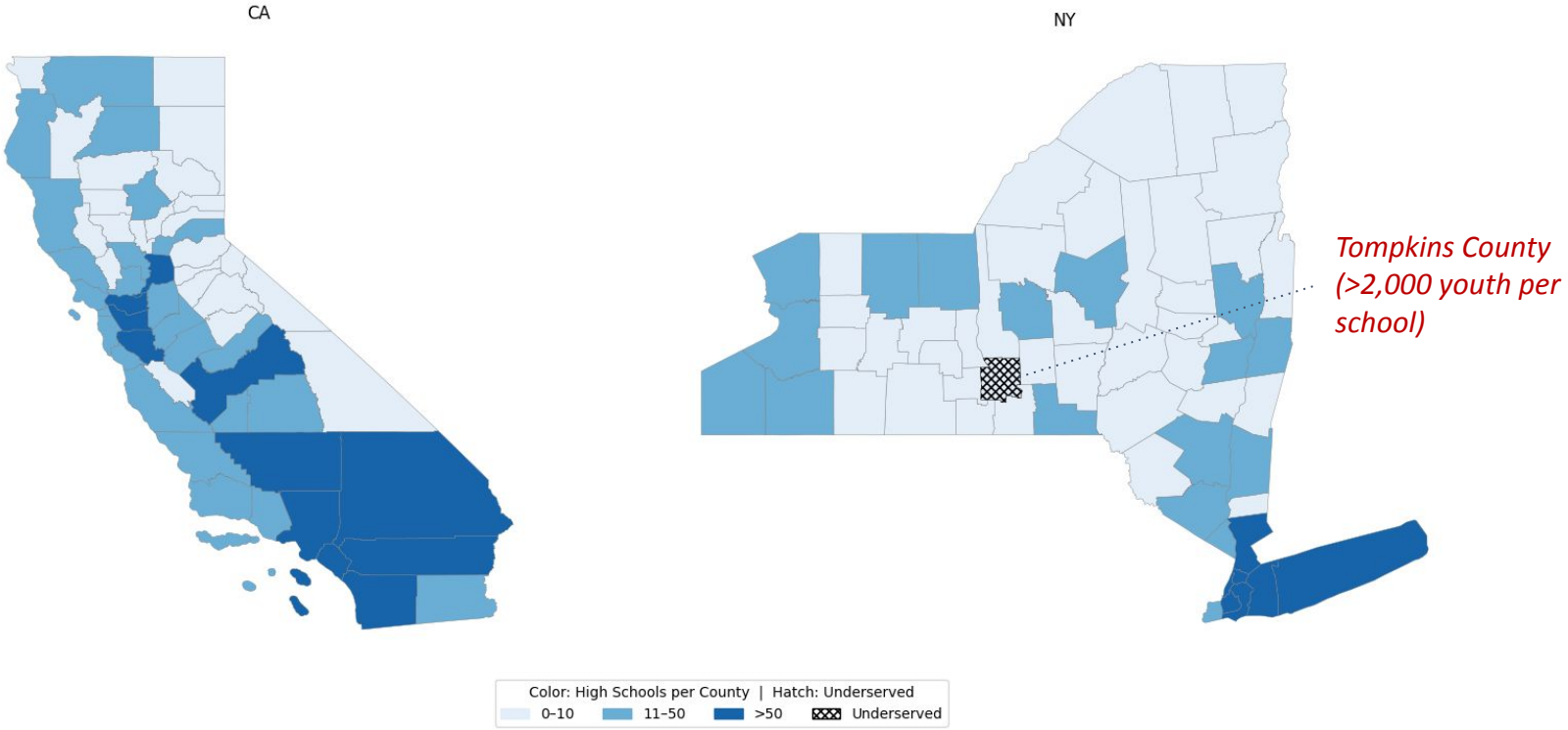
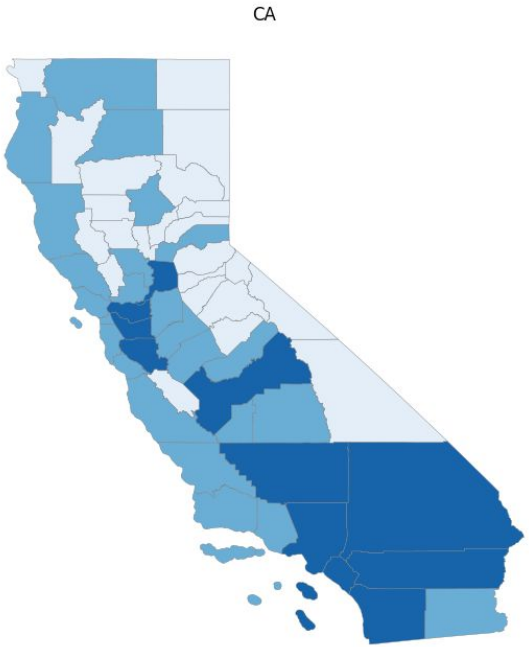


Figure 1. County-level distribution of high schools in California and New York State

County-Level School Distribution



California State

1,892

high schools statewide

58 counties total

39.7% have 0–10 schools

41.4% have 11–50 schools

19.0% have >50 schools

Top county: Los Angeles (372)

Top 5 counties = 42.3% of total

Figure 1. County-level distribution of high schools in California and New York State

County-Level School Distribution

New York State

999

high schools statewide

62 counties total

61.3% have 0–10 schools

27.4% have 11–50 schools

11.3% have >50 schools

Top county: Kings (97)

Top 5 counties = 38.3% of total

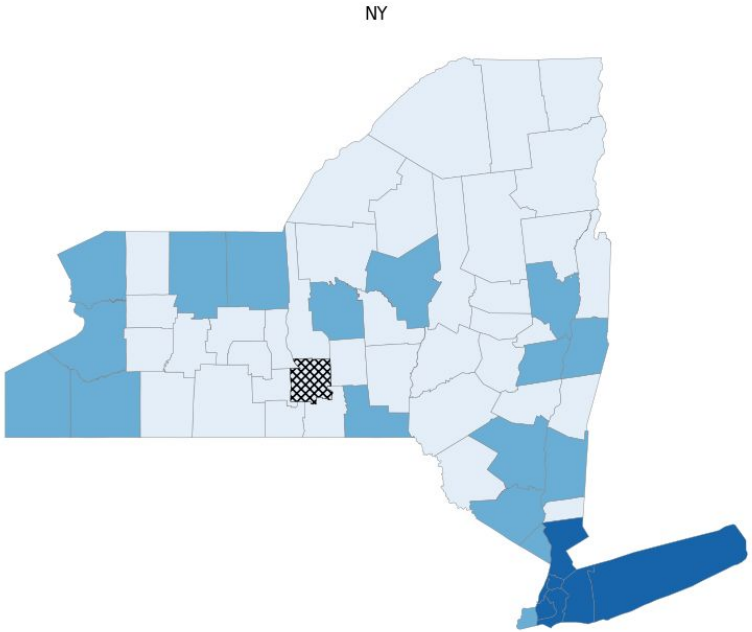


Figure 1. County-level distribution of high schools in California and New York State

County-Level: Key Numbers

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Block Group Accessibility (20km)

Key insight: Both states look similar in aggregate — but their spatial patterns are completely different.

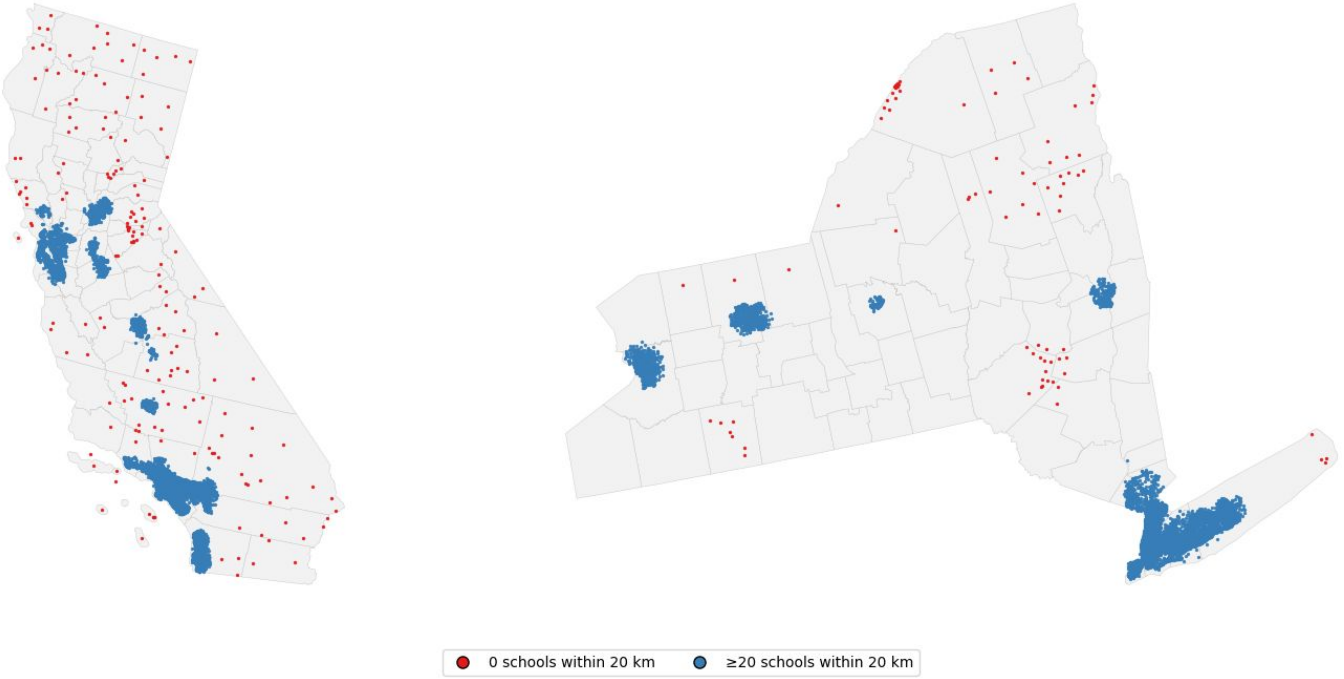
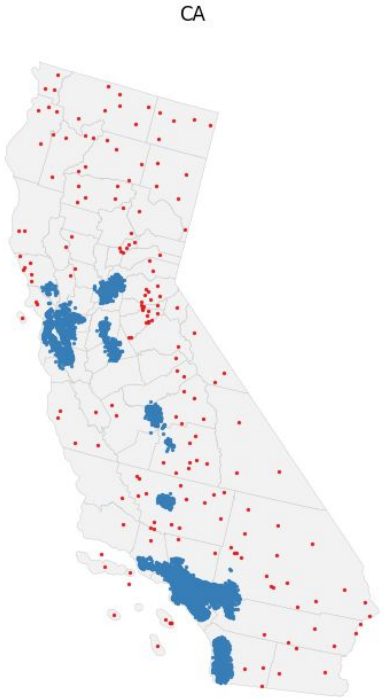


Figure 2. High school access extremes (20km). Red = 0 schools within reach; Blue = ≥20 schools within reach.

Block Group Accessibility (20km)



Within 20 km, 74.5% of California's BGs (72.4% of the population) have access to 20 or more high schools, while 0.7% (0.5%) have none.

California's zero-school BGs are more widely and evenly dispersed.

Figure 2. High school access extremes (20km). Red = 0 schools within reach; Blue = ≥20 schools within reach.

Block Group Accessibility (20km)

In New York, 69.3% of BGs (65.3% of the population) have over 20, and 0.5% (0.3%) have none.

New York's zero-school BGs are concentrated in a handful of counties.

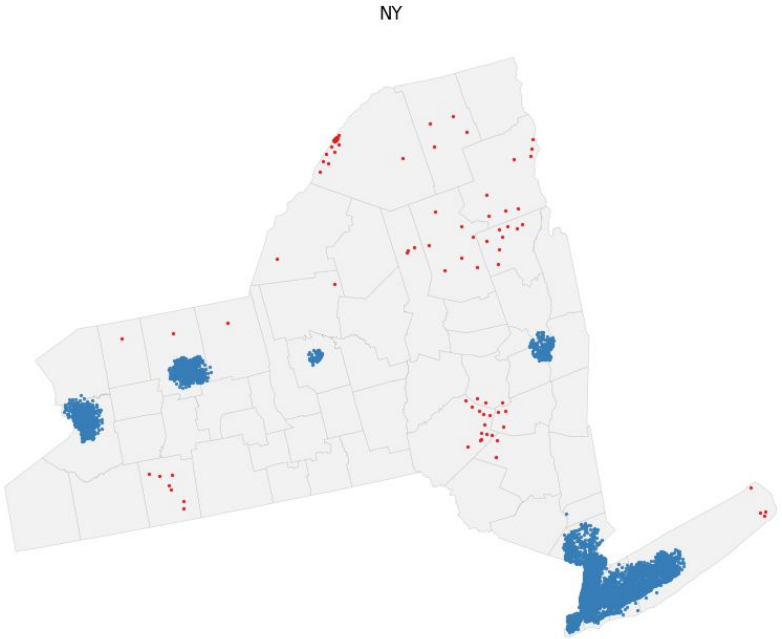


Figure 2. High school access extremes (20km). Red = 0 schools within reach; Blue = ≥20 schools within reach.



Takeaway

- GIS reveals the spatial dimension.
- CA’s educational deserts are scattered; NY’s are clustered → needs different policy

Limitations & Future Directions

Euclidean vs. Network Distance
We used 20km straight-line buffers, not actual road networks or transit routes. Real commute times may differ substantially, especially in rural areas.

Threshold Sensitivity
The 20km radius and 2,000-youth cutoff are exploratory choices. Different thresholds could shift which areas appear underserved.

School Quality Not Captured
We mapped location only — not capacity, quality, specialization, or admissions selectivity. Proximity ≠ true access.

Cross-Sectional Snapshot
One year of data (2021–22). Cannot capture school openings, closures, or demographic shifts over time.

Two-State Scope
CA and NY are diverse but not representative of all U.S. contexts. Future work could extend to more states or use network-based travel analysis.

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Thank You

Questions & Discussion

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